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The Granular Origins of Aggregate Fluctuations

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1 Big picture

- **Question.** Can idiosyncratic shocks to individual large firms explain a meaningful share of aggregate fluctuations, rather than averaging out as in the standard representative-firm intuition?
- **Core idea.** If firm sizes are fat-tailed, especially close to Zipf’s law, the central limit theorem works very slowly. A small number of large firms are such large shares of GDP that firm-specific shocks need not wash out in the aggregate.
- **Main theoretical mechanism.** In a standard equal-size economy, aggregate volatility falls at rate $1/\sqrt{N}$. In a granular economy with a fat-tailed firm-size distribution, aggregate volatility falls much more slowly; under Zipf’s law, it decays roughly as $1/\log N$.
- **Main empirical object.** The paper constructs a *granular residual*: a sales-share-weighted sum of idiosyncratic productivity-growth shocks for the largest 100 non-oil, non-financial firms in Compustat.
- **Main empirical result.** The granular residual explains about one-third of U.S. GDP growth fluctuations, and its explanatory power remains after controlling for industry shocks, oil shocks, monetary shocks, interest rates, and the term spread.
- **Narrative evidence.** Major years in the granular residual can be traced to concrete firm events, such as U.S. Steel strikes, GM price wars, Ford strikes, IBM’s PC launch, GE’s move into finance, and Walmart’s growth.
- **Economic takeaway.** Aggregate shocks may have microeconomic names. Business cycles need not originate only in economy-wide shocks; they can also originate in the incompressible “grains” of large-firm activity.

2 Incremental contribution of the paper

- **Microfoundation for aggregate shocks.** The paper gives a tractable microfoundation for what macroeconomists often treat as exogenous aggregate productivity shocks. Instead of taking aggregate shocks as primitive, it derives them from firm-level shocks weighted by firm size.
- **Fat-tailed aggregation theorem.** The paper clarifies why the usual diversification logic fails when firm sizes have a power-law tail. This resolves and reframes earlier debates about whether sectoral or firm-specific shocks can survive aggregation.
- **Granular residual as a new empirical object.** The paper constructs a practical measure of large-firm idiosyncratic shocks using Compustat sales and employment data. This gives researchers a concrete aggregate predictor based on identifiable firms.
- **Bridge between firm dynamics and macro fluctuations.** The analysis links Gibrat-style firm growth, Zipf’s law in firm size, Hulten’s theorem, and business-cycle volatility in one unified framework.
- **Quantitative calibration.** The paper shows that the magnitude of large-firm idiosyncratic volatility is in the right range to matter for business-cycle volatility once macro amplification mechanisms are allowed.
- **Empirical challenge to representative-agent macro.** The results suggest that macroeconomic fluctuations cannot always be understood by focusing only on aggregate variables. Large-firm histories may contain important information about GDP dynamics.
- **Narrative validation.** Unlike many purely statistical factor models, the paper connects residual movements to actual historical events at identifiable firms, making the mechanism economically interpretable.
- **Research program.** The paper creates a foundation for granular approaches to trade, stock returns, banking shocks, productivity, business groups, supply networks, and national output volatility.

3 Data and measurement

3.1 Firm universe and exclusion rules

- The empirical analysis uses annual U.S. Compustat data from 1951 to 2008.
- In each year, the granular residual is based on the largest 100 firms by previous-year sales that have valid sales and employment data.
- Oil, energy, and finance firms are excluded because sales and employment are poor proxies for physical output and productivity in those sectors.
- The focus on previous-year size avoids mechanical selection on current shocks.

Interpretation: The paper wants firm-level productivity shocks that are plausibly idiosyncratic. Excluding oil and finance reduces contamination from commodity-price and accounting-specific movements.

3.2 Productivity proxy

The firm-level productivity proxy is log sales per employee:

$$z_{it} = \ln \left(\frac{\text{sales of firm } i \text{ in year } t}{\text{employees of firm } i \text{ in year } t} \right).$$

Firm productivity growth is

$$g_{it} = z_{it} - z_{i,t-1}.$$

Interpretation: Sales per employee is imperfect, but it is broadly available and can be computed consistently across a long historical sample. The paper explicitly treats this as a first-pass granular residual, not a fully purified TFP measure.

3.3 Idiosyncratic firm shock

The ideal idiosyncratic shock is the residual from a decomposition such as

$$g_{it} = \beta' X_{it} + \varepsilon_{it},$$

where X_{it} contains factors that capture aggregate or common movements. The empirical analysis uses several ways to remove common components, including simple demeaning and industry demeaning.

Interpretation: The reflection problem is central: large firms may move because of aggregate shocks, or aggregate outcomes may move because of large firms. The granular residual is an attempt to isolate the latter channel.

3.4 Granular residual

The core empirical measure is a size-weighted sum of idiosyncratic shocks:

$$\Gamma_t = \sum_{i=1}^K \frac{S_{i,t-1}}{Y_{t-1}} \hat{\varepsilon}_{it}, \quad K = 100.$$

A simple industry-demeaned version uses

$$\hat{\varepsilon}_{it} = g_{it} - \bar{g}_{I(i),t},$$

where $\bar{g}_{I(i),t}$ is the average productivity growth of firms in the same three-digit SIC industry among the top 100.

Interpretation: A large-firm shock matters only if the firm is large. This is why the residual is weighted by the firm's lagged sales share of GDP.

3.5 Macroeconomic outcomes

- Main dependent variables are per capita GDP growth and Solow residual growth.
- Traditional predictors include oil shocks, Romer-Romer monetary shocks, interest rates, and the term spread.
- Narrative sections compare granular residual movements with historical firm events.

Interpretation: The empirical strategy asks whether the granular residual contains explanatory and predictive power beyond conventional macro controls.

4 Models and theoretical specifications

4.1 Islands economy

In the simple islands model, firm i produces output S_{it} . Aggregate output is

$$Y_t = \sum_{i=1}^N S_{it}.$$

Firm growth follows

$$\frac{\Delta S_{i,t+1}}{S_{it}} = \sigma_i \epsilon_{i,t+1},$$

where shocks are uncorrelated. Aggregate growth is a sales-share-weighted sum of firm shocks:

$$\frac{\Delta Y_{t+1}}{Y_t} = \sum_i \frac{S_{it}}{Y_t} \sigma_i \epsilon_{i,t+1}.$$

Interpretation: This equation is the granular idea in its simplest form: aggregate growth is a weighted average of micro shocks, and the weights are firm sizes.

4.2 Herfindahl volatility formula

If all firms have the same volatility $\sigma_i = \sigma$, aggregate volatility is

$$\sigma_{GDP} = \sigma h,$$

where

$$h = \left[\sum_i \left(\frac{S_i}{Y} \right)^2 \right]^{1/2}$$

is the square root of the sales Herfindahl.

Interpretation: In equal-size economies, $h = 1/\sqrt{N}$ and idiosyncratic shocks vanish. In concentrated economies, h can be economically large.

4.3 Proposition 1: equal-size benchmark

If firm sizes are small and not fat-tailed, aggregate volatility declines at the usual central-limit-theorem rate:

$$\sigma_{GDP} \propto \frac{1}{\sqrt{N}}.$$

Interpretation: This is the conventional diversification result. It explains why macroeconomists often ignore purely idiosyncratic firm shocks.

4.4 Proposition 2: fat-tailed firm sizes

If firm sizes follow a power-law distribution with exponent ζ , the aggregation rate changes. In the Zipf case $\zeta = 1$, aggregate volatility decays roughly as

$$\sigma_{GDP} \propto \frac{1}{\log N}.$$

Interpretation: This is the theoretical heart of the paper. When firm size is fat-tailed, there is no rapid averaging out. A handful of very large firms can dominate aggregate volatility.

4.5 Proposition 3: granular scaling and cross-country volatility

The paper extends the islands result to economies with productivity shocks and derives how volatility scales with country size. Under Zipf's law and Gibrat-style firm volatility, firm-level and country-level volatility can scale similarly.

Interpretation: This connects firm-size distributions to cross-country facts: smaller or less diversified countries can be more volatile because their output is more granular.

4.6 Hulten theorem connection

Hulten's theorem implies that, to first order, aggregate TFP changes can be expressed as Domar-weighted firm productivity shocks. The paper uses sales shares as empirical proxies for those weights.

Interpretation: This justifies using sales rather than value added in Figure 1 and in the granular residual.

4.7 Granular residual regressions

The main empirical specifications are

$$g_t^{GDP} = \alpha + \sum_{\ell=0}^L \beta_{\ell} \Gamma_{t-\ell} + u_t,$$
$$Solow_t = \alpha + \sum_{\ell=0}^L \beta_{\ell} \Gamma_{t-\ell} + u_t.$$

Robustness specifications add oil shocks, monetary shocks, interest rates, and the term spread.

Interpretation: The regressions quantify how much of aggregate volatility is statistically associated with the granular residual.

5 Identification and empirical design

5.1 Core challenge: the reflection problem

Large firms may move because of aggregate shocks, and aggregate shocks may move because of large firms. The paper cannot solve this perfectly with a randomized instrument, so it uses several empirical filters.

- It uses firm productivity growth rather than firm sales growth alone.
- It demeans firm growth by industry to remove sector-level shocks.
- It excludes oil and finance sectors where common shocks are especially problematic.
- It controls for traditional aggregate shocks in robustness regressions.
- It narrates individual large-firm events to assess plausibility.

Interpretation: The paper is transparent that the granular residual is provisional. Its contribution is to show that a plausible first-pass residual has large explanatory power.

5.2 Why top 100 firms?

- The theory says that the largest firms dominate the sales Herfindahl.
- Using the largest 100 firms captures the main granular force while keeping the measure interpretable.
- The top 100 are selected using lagged sales to avoid current-shock selection.

Interpretation: The granular view does not require every firm in the economy. The largest firms are enough to capture a large share of granular volatility.

5.3 Why narrative evidence matters

The narrative table and discussion connect large movements in the residual to identifiable events: strikes, model introductions, price wars, corporate restructurings, and major firm-level successes or failures.

Interpretation: This guards against interpreting the residual as a black-box factor. The granular residual has firm-level stories attached to it.

5.4 Limitations

- Sales per employee is not a fully purified TFP shock.
- Industry demeaning may not remove all common shocks.
- Compustat coverage and firm definitions change over time.
- Mergers, acquisitions, and corporate restructuring can contaminate firm growth measures.
- OLS regressions with annual data have small samples.

Interpretation: The findings should be read as a strong empirical opening rather than the final measurement of granular shocks.

6 Section-by-section study guide

- **Introduction.** Motivates the granular hypothesis: large firm shocks can become aggregate shocks because large firms are big shares of GDP.
- **Theory: islands economy.** Derives the relationship between aggregate volatility, firm volatility, and the sales Herfindahl.
- **Power-law firm-size section.** Shows that with a fat-tailed firm-size distribution, the central limit theorem does not average away idiosyncratic shocks quickly.
- **Calibration.** Uses estimates of firm-level productivity volatility and the sales Herfindahl to show that granular volatility is of business-cycle magnitude.
- **Granular residual construction.** Builds a size-weighted measure of top-firm idiosyncratic productivity movements.
- **Regression evidence.** Shows the granular residual explains GDP growth and Solow residual growth.
- **Narrative evidence.** Links residual episodes to identifiable firm-level events.
- **Conclusion.** Describes implications for macro research, VAR shocks, firm-level narratives, and other aggregates.

- The adjusted R^2 for GDP growth reaches 0.346 with current and lagged residuals.
- Table II uses an industry-demeaned residual to control for three-digit SIC industry shocks.
- The industry-demeaned version has even higher explanatory power: adjusted R^2 reaches 0.477 for GDP growth.

Detailed interpretation:

- Table I is the baseline empirical test. It asks whether large-firm idiosyncratic movements are statistically related to aggregate output and TFP.
- Table II is stronger because it removes industry-wide movements inside the top-firm sample. The fact that explanatory power rises rather than disappears supports the granular interpretation.
- Both tables show positive and significant coefficients on current and lagged residuals, consistent with firm shocks affecting aggregate growth over more than one year.

Economic message: The granular residual is not just a noisy firm-level statistic. It explains a quantitatively large share of GDP and Solow residual movements.

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	GDP Growth _t		Solow _t	
(Intercept)	0.018** (0.0026)	0.017** (0.0025)	0.011** (0.002)	0.01** (0.0021)
I_t	1.8* (0.69)	2.5** (0.69)	2.1** (0.54)	2.3** (0.57)
I_{t-1}	2.6** (0.71)	2.9** (0.67)	1.2* (0.55)	1.3* (0.56)
I_{t-2}		2.1** (0.71)		0.65 (0.59)
N	56	55	56	55
R^2	0.266	0.382	0.261	0.281
Adj. R^2	0.239	0.346	0.233	0.239

^aFor the year $t = 1952$ to 2008, per capita GDP growth and the Solow residual are regressed on the granular residual I_t of the top 100 firms (equation (33)). The firms are the largest by sales of the previous year. Standard errors are given in parentheses.

(a) Table I: Explanatory power of granular residual.

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	GDP Growth _t		Solow _t	
(Intercept)	0.019** (0.0024)	0.017** (0.0022)	0.011** (0.0019)	0.011** (0.0019)
I_t	3.4** (0.86)	4.5** (0.82)	3.3** (0.68)	3.7** (0.72)
I_{t-1}	3.4** (0.82)	4.3** (0.78)	1.5* (0.65)	1.9** (0.68)
I_{t-2}		2.7** (0.79)		0.77 (0.69)
N	56	55	56	55
R^2	0.356	0.506	0.334	0.372
Adj. R^2	0.332	0.477	0.309	0.335

^aFor the year $t = 1952$ to 2008, per capita GDP growth and the Solow residual are regressed on the granular residual I_t of the top 100 firms (equation (34)), removing the industry-demeaned residual.

(b) Table II: Industry-demeaned granular residual.

7.3 Table III and Table IV: Granular residual versus traditional predictors

Read:

- Table III studies predictive power relative to lagged GDP growth, oil shocks, monetary shocks, interest rates, and the term spread.
- The granular residual provides incremental predictive power beyond traditional macro and financial predictors.
- Table IV studies explanatory power controlling for oil, monetary shocks, and interest-rate variables.
- In Table IV, the granular residual alone has adjusted R^2 around 0.477, and all variables together yield adjusted R^2 around 0.767.

Detailed interpretation:

- Table III is the forecasting table. It asks whether top-firm shocks contain information about next-period GDP growth beyond conventional predictors.
- Table IV is the robustness table. It shows that granular shocks and standard macro variables are complements rather than substitutes.
- The granular residual performs nearly as well as traditional variables together in some specifications, which is striking given its simple construction.

Economic message: Firm-level shocks are not just an ex post narrative. They contain macro-relevant information that complements oil, money, and financial variables.

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TABLE III
PREDICTIVE POWER OF THE GRANULAR RESIDUAL FOR TERM SPREAD,
OIL SHOCKS, AND MONETARY SHOCKS*

	1	2	3	4	5	6	7	8
(Intercept)	0.022** (0.0029)	0.02** (0.0029)	0.022** (0.0029)	0.026** (0.0057)	0.015 (0.0075)	0.015 (0.0079)	0.019** (0.0027)	0.021** (0.0073)
Oil _{t-1}	-0.00027* (0.00012)		-0.00024* (0.00012)		-8.7e-05 (0.00013)		-0.00017 (0.00012)	
Oil _{t-2}	-0.00018 (0.00012)		-0.00017 (0.00012)		-6.9e-05 (0.00012)		-0.00012 (0.00011)	
Monetary _{t-1}	-0.083 (0.057)	-0.08 (0.055)			-0.042 (0.055)		-0.051 (0.05)	
Monetary _{t-2}	-0.059 (0.057)	-0.038 (0.056)			-0.024 (0.054)		0.043 (0.053)	
r _{t-1}				-0.75** (0.2)	-0.6 (0.32)	-0.45 (0.37)	-0.41 (0.34)	
r _{t-2}				0.65** (0.19)	0.56 (0.32)	0.43 (0.37)	0.39 (0.34)	
Term spread _{t-1}					0.32 (0.6)	0.38 (0.64)	0.4 (0.58)	
Term spread _{t-2}					0.45 (0.47)	0.27 (0.54)	-0.38 (0.53)	
r _{t-1}							3.5** (0.96)	3.3** (1)
r _{t-2}							1.2 (0.92)	2.5* (0.97)

(a) Table III: Predictive power versus traditional factors.

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TABLE IV
EXPLANATORY POWER OF THE GRANULAR RESIDUAL FOR
OIL AND MONETARY SHOCKS, AND INTEREST RATES*

	1	2	3	4	5	6	7	8
(Intercept)	0.023** (0.003)	0.02** (0.0029)	0.022** (0.003)	0.017** (0.0022)	0.019** (0.0023)	0.016* (0.0065)	0.02** (0.005)	0.023** (0.0048)
Oil _t	-9.8e-05 (0.00011)		-8.3e-05 (0.00012)		-4.6e-05 (8.6e-05)		-7.9e-05 (7.5e-05)	
Oil _{t-1}	-0.00028* (0.00012)		-0.00026* (0.00012)		-0.00021* (8.8e-05)		-0.00019* (7.5e-05)	
Oil _{t-2}	-0.00019 (0.00012)		-0.00019 (0.00012)		-0.00012 (8.9e-05)		-4.3e-05 (6.8e-05)	
Monetary _t		-0.0088 (0.059)	-0.03 (0.058)		-0.057 (0.043)		-0.044 (0.032)	
Monetary _{t-1}		-0.08 (0.061)	-0.065 (0.059)		0.012 (0.047)		-0.013 (0.033)	
Monetary _{t-2}		-0.061 (0.059)	-0.048 (0.058)		0.031 (0.046)		0.095** (0.033)	
r _t				4.5** (0.82)	4.2** (0.88)		3.7** (0.69)	4** (0.66)
r _{t-1}				4.3** (0.78)	4.5** (0.85)		2.8** (0.71)	3.6** (0.68)
r _{t-2}				2.7** (0.79)	2.7** (0.8)		2.8** (0.69)	2.8** (0.63)
r _t						0.66* (0.26)	0.69** (0.2)	0.83** (0.19)

(b) Table IV: Controls for oil, money, and rates.

7.4 Table V: Narrative

Read:

- Table V lists major granular episodes from 1952 through 2002.
- It reports firm share of GDP, labor productivity growth, demeaned growth, the granular residual contribution, direct share of GDP growth, imputed share, and a brief explanation.
- Examples include U.S. Steel strikes, GM model booms and price wars, Ford strikes, IBM's PC launch, GE's move into finance and high-tech, AT&T's spin-off, and Walmart's success.
- Some entries show direct shares of GDP growth that are large relative to annual aggregate fluctuations.

Detailed interpretation:

- Table V is a diagnostic table rather than a regression table. It asks whether the residual is economically interpretable.
- The table also shows why the granular approach is different from an anonymous factor model: the shocks have firm names and stories.
- The GM 1970 strike example is especially important because it illustrates how a single firm can account for a meaningful share of an aggregate GDP movement.
- The table's later entries show that the mechanism is not only about manufacturing strikes; it also includes information technology, finance, retail, and telecommunications.

Economic message: Granular shocks are historically concrete. The paper’s macro residual can be connected to firm-specific events that economists can study directly.

TABLE V
NARRATIVE^a

Year	Firm	Share of GDP $\frac{S_{i,t-1}}{Y_{t-1}}$ in %	Labor Prod. Growth $\Delta \ln S_{i,t}$ in %	Demeaned Growth $\bar{g}_{i,t} - \bar{g}_{i,t}$ in %	Gran. Res. $\bar{g}_{i,t}$ in %	Share of GR $\frac{f_{i,t}}{F_t}$	Direct Share of $g_{Y,t}$ $\frac{f_{i,t}}{F_t}$	Imputed Share of $g_{Y,t}$ $\frac{\mu f_{i,t}}{F_t}$	Brief Explanation
1952	U.S. Steel	1.03	[-10.75 -13.10, -2.35]	-3.56	-0.037	0.061	-0.810	-2.430	Strike
1953	U.S. Steel	0.87	17.06 [19.51, 2.45]	5.86	0.051	3.985	0.060	0.180	Rebound from strike
1955	GM	2.58	14.00 [21.89, 7.88]	17.84	0.461	0.808	0.142	0.426	Boom in car production: New models and price war
1956	Ford	1.35	-20.95 [-21.96, -1.01]	-20.72	-0.270	0.407	0.145	0.435	End of price war
1956	GM	3.00	-13.55 [-17.61, -4.06]	-13.32	-0.400	0.603	0.215	0.645	End of price war
1957	GM	2.47	0.36 [-1.50, -1.85]	-12.38	-0.305	2.201	0.167	0.501	End of price war (aftermath)
1961	Ford	1.00	25.12 [23.64, -1.48]	27.03	0.199	4.131	-0.147	-0.441	Success of compact Falcon (rebound from Edsel failure)
1965	GM	2.56	7.45 [18.06, 10.61]	11.10	0.284	0.600	0.092	0.276	Boom in new-car sales
1967	Ford	1.55	-19.84 [-18.23, 1.61]	-14.91	-0.232	2.461	0.379	1.137	Strike
1970	GM	2.47	-17.85 [-31.06, -13.20]	-20.52	-0.493	0.757	0.165	0.495	Strike
1971	GM	1.81	25.58 [36.15, 10.57]	23.35	0.361	0.516	7.344	22.032	Rebound from strike

(Continued)

(a) Table V: Narrative, part 1.

TABLE V—Continued

Year	Firm	Share of GDP $\frac{S_{i,t-1}}{Y_{t-1}}$ in %	Labor Prod. Growth $\Delta \ln S_{i,t}$ in %	Demeaned Growth $\bar{g}_{i,t} - \bar{g}_{i,t}$ in %	Gran. Res. $\bar{g}_{i,t}$ in %	Share of GR $\frac{f_{i,t}}{F_t}$	Direct Share of $g_{Y,t}$ $\frac{f_{i,t}}{F_t}$	Imputed Share of $g_{Y,t}$ $\frac{\mu f_{i,t}}{F_t}$	Brief Explanation
1972	Chrysler	0.71	16.76 [15.64, -1.13]	17.80	0.126	0.234	0.058	0.174	Rush of sales for subcompacts (Dodge Dart and Plymouth Valiant)
1972	Ford	1.46	14.18 [16.36, 2.18]	15.22	0.222	0.411	0.103	0.309	Rush of sales for subcompacts (Ford Pinto)
1974	GM	2.59	-11.31 [-21.28, -9.97]	-15.23	-0.394	0.913	0.115	0.345	Cars with poor gas mileage hit by higher oil price
1983	IBM ^b	1.06	10.46 [11.76, 1.29]	10.52	0.111	0.177	0.071	0.213	Launch of the IBM PC
1987	GE ^b	0.79	25.62 [8.35, -17.29]	21.46	0.158	1.110	0.357	1.071	Moving out of manufacturing and into finance and high-tech
1988	GE ^b	0.83	21.42 [20.08, -1.33]	16.55	0.137	0.441	0.117	0.351	Moving out of manufacturing and into finance and high-tech
1996	AT&T	1.08	38.97 [-44.11, -83.08]	32.45	0.215	0.471	0.446	1.338	Spin-off of NCR and Lucent
2000	GE	1.20	20.56 [12.29, -8.27]	33.04	0.239	9.954	0.468	1.404	Sales topped \$111bn, expansion of GE Medical Systems
2002	Walmart	2.16	8.61 [9.83, 1.22]	6.39	0.138	3.219	-0.099	-0.297	Success of lean distribution model

^aGE and GM are General Electric and General Motors, respectively. For each firm i , $\bar{g}_{i,t}$, $\Delta \ln S_{i,t}$, and $\Delta \ln L_{i,t}$ denote productivity, sales, and employment growth rates, respectively. $\bar{g}_{i,t} - \bar{g}_{i,t}$ denotes industry-demeaned growth, and $S_{i,t-1}/Y_{t-1}$ is the sales share of GDP. The firm granular residual is $f_{i,t} = \frac{S_{i,t} - (Y_{i,t} - \bar{g}_{i,t})}{Y_{i,t}}$, and F_t/f_t is the respective share of the granular residual. $f_{i,t}/F_t$ is the direct share of the firm shock on demeaned GDP growth. The full share would be equal to $\mu f_{i,t}/F_t$, where $\mu = 3$ is the typical productivity multiplier estimated from Tables I and II.

^bThere is just one firm in this industry in the top 100, hence $\bar{g}_{i,t}$ was replaced by $\bar{g}_t$.

(b) Table V: Narrative, part 2.

8 Theoretical result blocks

8.1 Result block 1: Why the central limit theorem fails

- Equal-size firms generate standard $1/\sqrt{N}$ diversification.
- Power-law firm-size distributions generate much slower diversification.
- Under Zipf’s law, firm shares are so unequal that the largest firms continue to dominate aggregate volatility.

Interpretation: The paper’s core theory is not that every firm matters. It is that the largest firms matter because the size distribution is extremely skewed.

8.2 Result block 2: Sales Herfindahl as sufficient statistic

- Aggregate volatility is proportional to firm volatility times the sales Herfindahl.
- The sales Herfindahl is observable, so the theory has direct empirical discipline.
- In concentrated economies or small countries, the Herfindahl is larger and granular volatility is stronger.

Interpretation: The Herfindahl formula provides a transparent bridge from micro size distributions to macro volatility.

8.3 Result block 3: Hulten theorem and Domar weights

- Hulten’s theorem motivates weighting firm productivity shocks by sales shares rather than purely by value added.
- Large firms with large sales are important because their shocks have large first-order aggregate effects.
- This justifies the empirical construction of the granular residual.

Interpretation: The paper is careful to connect the residual to growth accounting logic rather than simply summing large-firm news.

8.4 Result block 4: Granular residual as macro predictor

- The residual is computed from top-firm shocks and lagged size weights.
- It explains and forecasts GDP growth beyond traditional macro indicators.
- It can be used as a real-side shock series analogous to monetary or oil shock series.

Interpretation: The empirical contribution is not just that large firms are large. It is that their idiosyncratic movements have measurable macro predictive content.

9 Detailed result map

- **Concentration fact:** top 50 and top 100 firms are large shares of GDP.
- **Theoretical result:** fat-tailed firm sizes make idiosyncratic shocks survive aggregation.
- **Calibration result:** large-firm volatility times the sales Herfindahl is business-cycle relevant.
- **Empirical result:** the granular residual explains about one-third of U.S. GDP growth variation.
- **Robustness result:** the residual retains explanatory power after industry demeaning and conventional macro controls.
- **Forecasting result:** lagged granular shocks add predictive power beyond traditional variables.
- **Narrative result:** major residual movements correspond to concrete firm events.

10 Short summary

- Gabaix argues that aggregate fluctuations partly originate in idiosyncratic shocks to very large firms.
- Standard diversification logic fails when firm sizes are fat-tailed.
- Under Zipf's law, aggregate volatility declines slowly, not at the usual $1/\sqrt{N}$ rate.
- The paper constructs a granular residual from the top 100 U.S. firms using sales weights and productivity-growth shocks.
- The residual explains a large share of GDP growth and Solow residual variation.
- It also adds forecasting power beyond oil, monetary, interest-rate, and term-spread variables.
- Historical episodes in the residual can be linked to identifiable firm events.
- The paper creates a new research program connecting micro firm dynamics to macro fluctuations.

11 Conclusion

11.1 Core conclusion

Gabaix shows that idiosyncratic firm-level shocks can be aggregate shocks when firm sizes are fat-tailed. In such an economy, the largest firms are so important that their productivity shocks do not diversify away. The granular residual constructed from the top 100 firms explains a large fraction of aggregate fluctuations.

11.2 Economic intuition

In a world of many equal-size firms, an idiosyncratic shock to any one firm is negligible. In a granular economy, a few firms represent large shares of activity. A strike, product innovation, logistics failure, price war, or managerial mistake at a large firm can move aggregate output directly and through general equilibrium spillovers.

11.3 Incremental contribution revisited

The paper's incremental contribution is to turn the distribution of firm sizes into a macroeconomic state variable. It combines Zipf's law, Hulten-style growth accounting, a firm-level residual, and macro regressions to show that the microeconomic composition of GDP matters for business-cycle analysis.

11.4 Implications for macroeconomics

The granular hypothesis implies that macroeconomists should study large firms, not only aggregate shocks. Firm-level innovations, strikes, restructurings, and failures can be inputs to VARs and impulse-response exercises, much like oil or monetary shocks.

11.5 Limitations

The paper is explicit that measuring idiosyncratic firm shocks is difficult. Sales per employee is an imperfect productivity proxy, common shocks are hard to remove, and firm events can be entangled with aggregate conditions. Still, the explanatory power of the residual suggests that the granular approach is too large to ignore.

11.6 Takeaway

The central takeaway is that macro volatility can have micro names. Aggregate fluctuations may be partly the sum of shocks to Walmart, GM, Ford, IBM, GE, and other large firms. This changes how researchers should think about the origins and propagation of business cycles.